Botany short course 2026

Introduction to British Botany

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Overview

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- Yet the flora punches above its weight: wide range of habitats, strong Atlantic influence, and centuries of botanical recording
- This lecture: how many plants, where they come from, the families they belong to, and how to find out more

How many species?

- Britain and Ireland have approximately **1,700 native vascular plant species**
- Including all vascular plants (native + naturalised + common aliens): **3500+ taxa** recorded (Stace, 4th ed.)
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- Flowering plants (angiosperms): 1,600 native spp.
- Ferns & allies (pteridophytes): 80 native spp.
- Conifers (gymnosperms): 3 native spp. (*Pinus sylvestris*, *Juniperus communis*, *Taxus baccata*)

Native, archaeophyte, or neophyte?

- **Native** — present before human influence; arrived naturally after the last glaciation
- **Archaeophyte** — introduced by humans **before** 1500 CE (often arable weeds arriving with agriculture)
- Examples: *Papaver rhoeas* (common poppy), *Centaurea cyanus* (cornflower)
- **Neophyte** — introduced **after** 1500 CE (post-Columbian exchange, horticulture, trade)
- Examples: *Fallopia japonica* (Japanese knotweed), *Rhododendron ponticum*, *Impatiens glandulifera*
- **Casual alien** — appears but does not persist without repeated introduction

Alien and invasive species

- Invasion biology distinguishes **naturalised** (self-sustaining) from **invasive** (spreading, causing harm)
- Key invasive species in Britain:
- *Fallopia japonica* — Japanese knotweed; dense monoculture, structural damage
- *Impatiens glandulifera* — Himalayan balsam; riverbanks, shades out natives
- *Rhododendron ponticum* — woodland understorey, toxic to stock
- *Crassula helmsii* — New Zealand pygmyweed; aquatic habitats

How do the aliens get here?

- Pathways: horticulture, agriculture, forestry, ballast water, contaminated soil, bird seed
- GB Non-Native Species Secretariat (GBNNSS) maintains the risk register

Diversity: major families

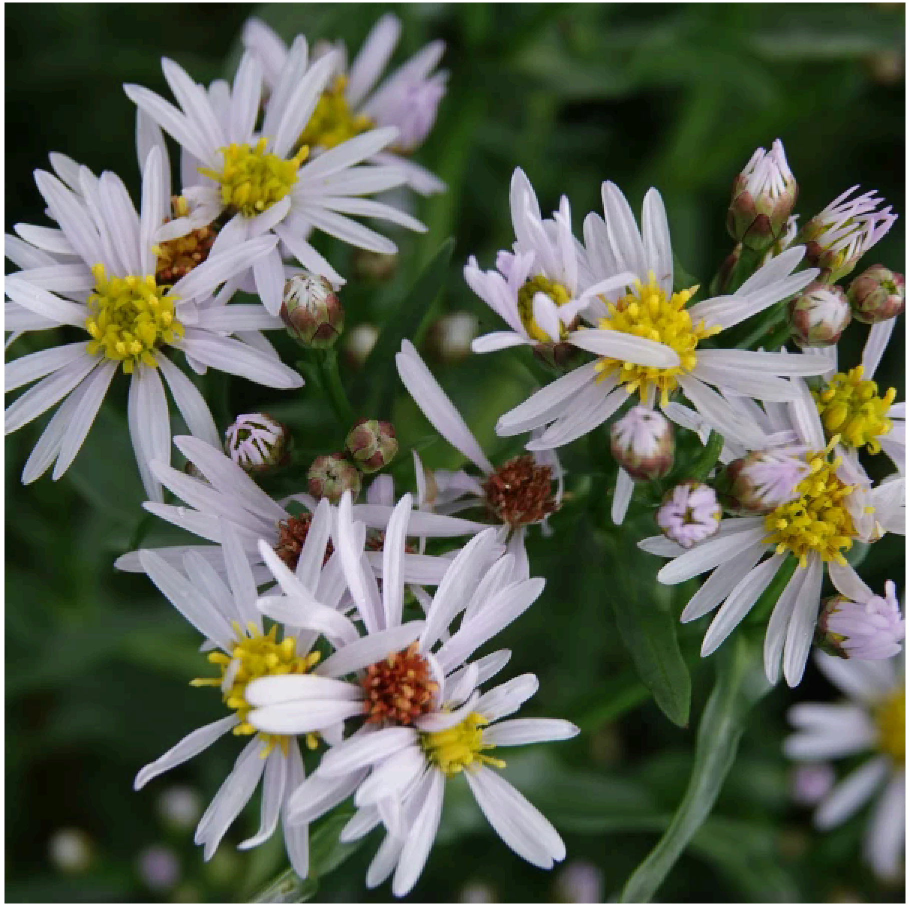
Asteraceae (230 spp.)
Rosaceae (130 spp.)
Brassicaceae (110 spp.)
Cyperaceae (120 spp.)

Poaceae (170 spp.)
Fabaceae (80 spp.)
Lamiaceae (70 spp.)
Apiaceae (70 spp.)
Ranunculaceae (65 spp.)
Caryophyllaceae (60 spp.)
Orchidaceae (52 spp.)

Although >3000 species including aliens, only ~180 **families**!

Diversity: major families

Asteraceae — daisy family — ~230 spp. in Britain



- Composite flowerhead (capitulum) made of many tiny florets
- Ray florets (strap-shaped) surround disc florets (tubular)
- Example: *Tripolium pannonicum* (sea aster) — saltmarsh specialist
- Key genera: *Senecio*, *Cirsium*, *Centaurea*, *Taraxacum*, *Hieracium*

Diversity: major families

Rosaceae — rose family — ~130 spp. in Britain

- 5 petals, many stamens, inferior or superior ovary depending on genus
- Includes trees, shrubs, and herbs — enormous ecological breadth
- Example: *Rosa canina* (dog rose) — hedgerow classic; hips rich in vitamin C
- Key genera: *Rosa*, *Rubus*, *Prunus*, *Sorbus*, *Potentilla*, *Agrimonia*

Diversity: major families

Brassicaceae — cabbage family — ~110 spp. in Britain

- Diagnostic: 4 petals in a cross, 6 stamens (4 long + 2 short), silique fruit
- Many archaeophyte arable weeds; also coastal and upland specialists
- Example: *Cakile maritima* (sea rocket) — strandline pioneer
- Key genera: *Cardamine*, *Arabidopsis*, *Rorippa*, *Draba*, *Raphanus*

Diversity: major families

Fabaceae — pea family — ~80 spp. in Britain

- Zygomorphic flowers; fruit a legume (pod)
- Root nodules with nitrogen-fixing *Rhizobium* bacteria
- Example: *Lathyrus linifolius* (bitter-vetch) — upland grassland and heath
- Key genera: *Trifolium*, *Vicia*, *Lathyrus*, *Lotus*, *Ulex*, *Genista*

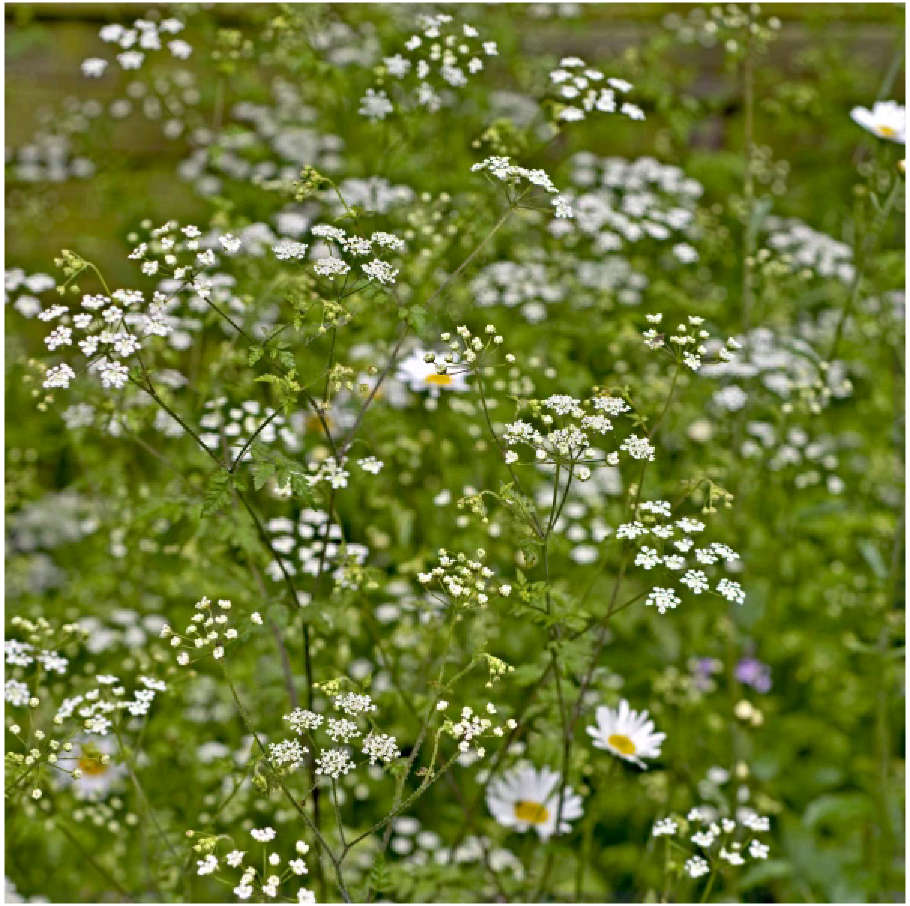
Diversity: major families

Lamiaceae — mint family — ~70 spp. in Britain

- Square stems, opposite leaves, bilabiate (2-lipped) flowers
- Many aromatic — volatile oils in glands on leaves
- Example: *Lamium purpureum* (red deadnettle) — common weed of disturbed ground
- Key genera: *Lamium*, *Stachys*, *Prunella*, *Mentha*, *Thymus*, *Ajuga*

Diversity: major families

Apiaceae — carrot family — ~70 spp. in Britain



- Compound umbels; small 5-petalled flowers; often hollow stems
- Aromatic; many edible genera but also deadly (*Conium*, *Oenanthe*)
- Example: *Chaerophyllum temulentum* (rough chervil) — hedgerow annual
- Key genera: *Anthriscus*, *Conium*, *Heracleum*, *Daucus*, *Oenanthe*

Diversity: major families

Ranunculaceae — buttercup family — ~65 spp. in Britain

- Mostly actinomorphic; many free stamens and carpels (primitive traits)
- Aquatic and wetland specialists in genus *Ranunculus*
- Example: *Ranunculus lingua* (greater spearwort) — tall fen species
- Key genera: *Ranunculus*, *Caltha*, *Aquilegia*, *Clematis*, *Anemone*

Diversity: major families

Caryophyllaceae — pink family — ~60 spp. in Britain

- Opposite, entire leaves; swollen nodes; 5 petals often notched
- Calyx often tubular and prominently veined
- Example: *Silene viscaria* (sticky catchfly) — dry rocky grassland, local in Britain
- Key genera: *Silene*, *Cerastium*, *Stellaria*, *Arenaria*, *Dianthus*

Diversity: major families

Orchidaceae — orchid family — ~52 spp. in Britain

- Highly zygomorphic; labellum (lip) often elaborate
- Mycorrhizal dependency — seeds lack endosperm, require fungal germination
- Example: *Orchis mascula* (early-purple orchid) — ancient woodland indicator
- Key genera: *Dactylorhiza*, *Ophrys*, *Gymnadenia*, *Epipactis*, *Neottia*

Biogeographic elements

- British flora classified into **phytogeographic elements** (Preston & Hill 1997)
- Key elements relevant to our flora:
  - **Oceanic** — concentrated in W Britain; require mild, humid winters (*Hymenophyllum*, *Euphorbia hyberna*)
  - **Continental** — SE Britain; drought-tolerant, warm summers (*Medicago minima*, arable weeds)
  - **Arctic-alpine** — upland Scotland and Wales; cold-adapted relicts (*Dryas octopetala*, *Saxifraga oppositifolia*)
  - **Mediterranean** — S England coasts; near range edge (*Cistus*, *Orobanche* spp.)
- Climate change already shifting range margins northward and upward

Recording and mapping

- Britain has one of the world's best-documented floras — recording since the 16th century
- **Botanical Society of Britain and Ireland (BSBI)** — coordinates recording; Atlas 2020 is the current distribution atlas
- **NBN Atlas** — national biodiversity network; open data portal for occurrence records
- **iNaturalist / iRecord** — citizen science; useful but verify IDs before trusting
- Recording unit: the **hectad** (10 × 10 km OS grid square); 2,900 hectads in Britain & Ireland
- Species distribution maps on BSBI.org by hectad

Key resources: field guides and floras

Floras (identification keys)

- **Stace, C.A.** — *New Flora of the British Isles*, 4th ed. (2019). The standard reference; comprehensive dichotomous keys.
- **Poland, J. & Clement, E.** — *The Vegetative Key to the British Flora* (2020). Identify by vegetative characters alone — invaluable when flowers are absent.
- **Rose, F. & O'Reilly, C.** — *The Wildflower Key*, revised ed. (2006). Accessible beginner flora; illustrated dichotomous keys.

Field guides (visual)

- **Blamey, M., Fitter, R. & Fitter, A.** — *Collins Wildflower Guide* (2013). Excellent illustrations; covers 1,400 spp.
- **Streeter, D. et al.** — *Collins Wildflower Field Guide* (2016). Photographic; very portable.

Choosing the right book

Situation	Best choice	Why
No flower present	Poland & Clement	Keys vegetative characters
Beginner, flower present	Rose	Simple keys, clear illustrations
Comprehensive determination	Stace	Full keys, all taxa
Quick visual check	Collins Wildflower Guide	Fast to scan illustrations
Data / distribution	BSBI Atlas 2020 / NBN	Maps and occurrence data

Aims of the course

- Have fun!
- Learn more about plants, get stuck in the detail, or beauty
- Learn plant **families**
- Go further if you can, but not a priority